FACULTY OF COMPUTING
UTM Johor Bahru

SCST1123: MATHEMATICS FOR SOFTWARE ENGINEERING
20242025/SEMESTER 1

ASSIGNMENT #4b (Finite State Machine Project)

Finite State Machine Project Instructions

Objective:
This project aims to deepen your understanding of finite state machines (FSMs) by analyzing and presenting a specific example of your choice.

Instructions:

1. **Select an FSM Example:**
   - Choose a real-world example of a finite state machine. Examples could include a vending machine, traffic light controller, turnstile, or any other system that operates based on states and transitions.
2. **Explain the FSM:**
   - Provide a clear and concise explanation of how the chosen FSM works.
   - Describe the states, inputs, outputs, and the logic governing the transitions between states.
3. **Create a Transition Table:**
   - Construct a transition table to represent the FSM.
   - The table should include all possible states, inputs, and the resulting next states. If applicable, include outputs.
4. **Draw a State Transition Diagram:**
   - Use appropriate symbols to visually represent the FSM.
   - The diagram should include all states, labelled transitions, and clearly indicate the start and any final states.
5. **Prepare a Presentation Recording:**
   - Create a presentation summarizing your project, including:
     - An introduction to your chosen FSM.
     - Explanation of the transition table and state transition diagram.
     - Discussion of how the FSM operates and its practical significance.
   - Record your presentation as a video or voice-over slideshow. The presentation should be **3-5 minutes long**.
6. **Document Your Work:**
   - Write a detailed report containing your explanation, the transition table, and the state transition diagram.
   - Ensure your report is neatly formatted and free of grammatical errors.

7. **Submission Guidelines:**
    - Save your report as a PDF file.
    - Save your presentation recording as an MP4 file or provide a link to an online version (e.g., YouTube, Google Drive).
    - Submit both files to the e-learning platform before the deadline: **Friday, 24/01/2025, 5:00 PM**.

Evaluation Criteria:
Your project will be assessed based on the following:

- Clarity and accuracy of the explanation (25%)
- Correctness and completeness of the transition table (25%)
- Accuracy and design of the state transition diagram (25%)
- Quality and delivery of the presentation recording (15%)
- Presentation and adherence to instructions (10%)

Note: Plagiarism is strictly prohibited. Ensure your work is original and properly cites any external references.